

1 Osnove teorije

1.1 BCS teorija pri končni temperaturi

BCS model, kjer je $\xi_{\mathbf{k},\sigma} = \epsilon_{\mathbf{k},\sigma} - E_F$ in $k := (\mathbf{k}, \uparrow)$, $-k := (-\mathbf{k}, \downarrow)$:

$$H = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - V \sum_{k,k'} c_k^\dagger c_{-k}^\dagger c_{-k'} c_{k'} \quad (1)$$

Naredimo približek povprečnega polja, definiramo $b_k = c_k c_{-k}$ in $b_k = \langle b_k \rangle + \delta b_k = F_k + \delta b_k$, sledi (do $\mathcal{O}(\delta b_k^2)$)

$$b_k^\dagger b_{k'} \simeq F_k^* F_{k'} + F_k^* \delta b_{k'} + F_{k'} \delta b_k^\dagger = F_k^* b_{k'} + F_{k'} b_k^\dagger - F_k^* F_{k'} = \langle b_k^\dagger \rangle b_{k'} + b_k^\dagger \langle b_{k'} \rangle - \langle b_k^\dagger \rangle \langle b_{k'} \rangle \quad (2)$$

Iz česar sledi:

$$H = \sum_k (c_k^\dagger \quad c_{-k}) \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^* & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \frac{|\Delta|^2}{V} + \sum_k \xi_k \quad (3)$$

Ob definiciji $\Delta_k = \sum_{k'} V(k - k') \langle c_{k'} c_{-k'} \rangle$, vzamemo še lokalno interakcijo, zato je $V(q) \simeq V$ in $\Delta_k \simeq \Delta$. Naredimo Bogoljubov rotacijo, privzamemo $u_k, v_k \in \mathbb{R}$ in simetrijo okoli E_F : $u_k = u_{-k}$ ter $v_k = v_{-k}$. Iz CCR sledi $u_k^2 + v_k^2 = 1$. Kvazidelci so definirani z rotacijo

$$\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix} \quad (4)$$

nov Hamiltonjan pa postane

$$H = \sum_k (\alpha_k^\dagger \quad \alpha_{-k}) \begin{pmatrix} \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) - \xi_k v_k^2 & -2\xi_k u_k v_k + \Delta_k u_k^2 - \Delta_k^* v_k^2 \\ -2\xi_k u_k v_k - \Delta_k v_k^2 + \Delta_k^* u_k^2 & \xi_k v_k^2 - \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix}, \quad (5)$$

kjer zahtevamo, da so izvendiagonalni členi enaki 0. Iz CCR zahteve lahko vzamemo parameterizacijo (tu sem vzel kompleksno nastavimo, da je $\Delta = \Delta^* e^{i2\phi_k}$):

$$v_k = \sin\left(\frac{\theta_k}{2}\right) e^{i\phi_k} \quad (6)$$

$$u_k = \cos\left(\frac{\theta_k}{2}\right) \quad (7)$$

$$u_k v_k = \cos\left(\frac{\theta_k}{2}\right) \sin\left(\frac{\theta_k}{2}\right) = \frac{1}{2} \sin \theta_k \quad (8)$$

$$u_k^2 - v_k^2 = \cos^2\left(\frac{\theta_k}{2}\right) - \sin^2\left(\frac{\theta_k}{2}\right) = \cos \theta_k \quad (9)$$

skupaj dobimo (od tu naprej $\Delta := |\Delta|$):

$$\xi_k \sin(\theta_k) = \Delta \cos(\theta_k) \quad (10)$$

kar fiksira

$$\tan(\theta_k) = \frac{|\Delta_k|}{\xi_k}. \quad (11)$$

Tako iz trikotnika preberemo

$$\sin \theta_k = \frac{|\Delta_k|}{E_k} = \frac{|\Delta_k|}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \quad (12)$$

$$\cos \theta_k = \frac{\xi_k}{E_k} = \frac{\xi_k}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \quad (13)$$

$$(14)$$

kjer je $E_k = \pm \sqrt{\xi_k^2 + |\Delta_k|^2}$ energija kvazidelcev:

$$H = \sum_k E_k \alpha_k^\dagger \alpha_k - E_k \alpha_{-k}^\dagger \alpha_{-k} + \xi_k + \frac{|\Delta|^2}{V}. \quad (15)$$

V termičnem stanju so diagonalni delci porazdeljeni po

$$\langle \alpha_k^\dagger \alpha_k \rangle = \frac{1}{1 + \exp(E_k/T)} = \frac{1}{2} - \frac{1}{2} \tanh\left(\frac{E_k}{2T}\right). \quad (16)$$

Uporabni sta zvezi $f(-E/T) = 1 - f(E/T)$ in $1 - 2f(E/T) = \tanh(E/(2T))$. V definiciji ureditvenega parametra je

$$\langle c_k c_{-k} \rangle = \left\langle \left(u_k \alpha_k - v_k \alpha_{-k}^\dagger \right) \left(u_k \alpha_{-k} + v_k \alpha_k^\dagger \right) \right\rangle = u_k v_k \left(1 - \langle \alpha_{-k}^\dagger \alpha_{-k} \rangle - \langle \alpha_k^\dagger \alpha_k \rangle \right) = \frac{1}{2} \sin \theta_k \tanh\left(\frac{\beta E_k}{2}\right) \quad (17)$$

iz česar sledi enačba za režo pri $T > 0$ (seštevamo po k okoli E_F):

$$\Delta = \frac{V}{2} \sum_k' \sin \theta_k \tanh\left(\frac{E_k}{2T}\right) = \frac{V\Delta}{2} \sum_k \frac{\tanh\left(\frac{1}{2T} \sqrt{\Delta^2 + \xi_k^2}\right)}{\sqrt{\Delta^2 + \xi_k^2}} \quad (18)$$

1.2 Minimalni model ekscitonskega izolatorja pri končni temperaturi

Pri izolatorju za minimalni model vzamemo dve orbitali (a, b) model tesne vezi v vsaki posebej in lokalno Coulombovo interakcijo na mestu j med elektronskima gostotama v posameznih pasovih

$$H = H_0 + H_V = \sum_{i,j} \left(\epsilon_a^{(0)} \delta_{ij} + t_{ij}^{(a)} \right) c_{a,i}^\dagger c_{a,j} + \sum_{i,j} \left(\epsilon_b^{(0)} \delta_{ij} + t_{ij}^{(b)} \right) c_{b,i}^\dagger c_{b,j} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (19)$$

$$= \sum_k \epsilon_{a,k} c_{a,k}^\dagger c_{a,k} + \sum_k \epsilon_{b,k} c_{b,k}^\dagger c_{b,k} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (20)$$

najprej diagonaliziramo vsako orbitalo posebej, kar da gole disperzije (vezavni energiji nastavimo na $\pm G/2$) (delamo v dveh dimenzijah in mrežna konstanta $a_0 = 1$):

$$\epsilon_{a(b),k} = \pm \frac{G}{2} - 2t_{a(b)} (\cos(k_x) + \cos(k_y)), \quad (21)$$

v tem primeru je najmanjša razdalja med pasovoma $G_0 = G - 4(t_a - t_b)$ oziroma $G_0 = G - 8t_a$, če je $t_b = -t_a$. Za interakcijo spet naredimo približek povprečnega polja:

$$c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \approx \langle c_{a,j}^\dagger c_{a,j} \rangle c_{b,j}^\dagger c_{b,j} + \langle c_{b,j}^\dagger c_{b,j} \rangle c_{a,j}^\dagger c_{a,j} - \langle c_{a,j}^\dagger c_{b,j} \rangle c_{b,j}^\dagger c_{a,j} - \langle c_{b,j}^\dagger c_{a,j} \rangle c_{a,j}^\dagger c_{b,j} + \text{konst.} \quad (22)$$

od tod sledi

$$H_V^{\text{MF}} = V n_b \sum_j c_{a,j}^\dagger c_{a,j} + V n_a \sum_j c_{b,j}^\dagger c_{b,j} - V \phi \sum_j c_{b,j}^\dagger c_{a,j} - V \phi^* \sum_j c_{a,j}^\dagger c_{b,j} + NV |\phi|^2 \quad (23)$$

kjer smo definirali:

$$n_a = \langle c_{a,j}^\dagger c_{a,j} \rangle = \frac{1}{N} \sum_k \langle c_{a,k}^\dagger c_{a,k} \rangle \quad (24)$$

$$n_b = \langle c_{b,j}^\dagger c_{b,j} \rangle = \frac{1}{N} \sum_k \langle c_{b,k}^\dagger c_{b,k} \rangle \quad (25)$$

$$\phi = \langle c_{a,j}^\dagger c_{b,j} \rangle = \frac{1}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle \quad (26)$$

$$\Delta = -\frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle, \quad \Delta^* = -\frac{V}{N} \sum_k \langle c_{b,k}^\dagger c_{a,k} \rangle \quad (27)$$

$$\Delta_k = \frac{1}{N} \sum_{k'} V(k - k') \langle a_{k'}^\dagger b_{k'} \rangle, \quad V(q) = \int e^{i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}) d\mathbf{r} \quad (28)$$

Uporabimo FT še na celotnem Hamiltonjanu

$$c_{a,j}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{i\mathbf{k} \cdot \mathbf{r}_j} c_{a,k}^\dagger \quad (29)$$

$$H^{\text{MF}} = \sum_k \left([\epsilon_{a,k} + V n_b] c_{a,k}^\dagger c_{a,k} + [\epsilon_{b,k} + V n_a] c_{b,k}^\dagger c_{b,k} + \Delta^* c_{a,k}^\dagger c_{b,k} + \Delta c_{b,k}^\dagger c_{a,k} \right) + \left(\frac{|\Delta|^2}{V} N - V N n_a n_b \right) \quad (30)$$

oziroma:

$$H^{\text{MF}} = \sum_k \begin{pmatrix} c_{a,k}^\dagger & c_{b,k}^\dagger \end{pmatrix} \begin{pmatrix} \epsilon_{a,k}^H & \Delta^* \\ \Delta & \epsilon_{b,k}^H \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} + \text{konst.}, \quad (31)$$

kjer definiramo Hartree energiji $\epsilon_{a(b),k}^H = \epsilon_{a(b),k} + V n_{b(a)}$. Transformacija v diagonalno / Bogoljubov bazo je podobna kot pri BCS, le da ne meša kreatorjev in anihilatorjev / ohranja število delcev:

$$\begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} \quad (32)$$

od tod po enakem postopku kot prej dobimo energiji kvazidelcev

$$E_{\alpha(\beta)}(k) = \frac{\epsilon_{a,k}^H + \epsilon_{b,k}^H}{2} \pm \sqrt{\left(\frac{\epsilon_{a,k}^H - \epsilon_{b,k}^H}{2} \right)^2 + |\Delta|^2} := \eta_k \pm \sqrt{\xi_k^2 + \Delta^2} \quad (33)$$

Za bookkeeping:

$$\left(\frac{\epsilon_a^H - \epsilon_b^H}{2} \right) \sin \theta_k + \Delta \cos \theta_k = 0 \implies \tan \theta_k = \frac{-2\Delta}{\epsilon_a^H - \epsilon_b^H} = \frac{-\Delta}{\xi_k} \quad (34)$$

$$u_k v_k = -\frac{\sin \theta_k}{2}. \quad (35)$$

Za samousklajeno enačbo zdaj potrebujemo

$$\langle c_{a,k}^\dagger c_{b,k} \rangle = \langle (u_k \alpha_k^\dagger + v_k \beta_k^\dagger)(-v_k \alpha_k + u_k \beta_k) \rangle = u_k v_k \left(\langle \beta_k^\dagger \beta_k \rangle - \langle \alpha_k^\dagger \alpha_k \rangle \right) = -\frac{1}{2} \sin \theta_k (f(E_\beta) - f(E_\alpha)) \quad (36)$$

od tod

$$\Delta = -\frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle = \frac{V}{N} \sum_k \frac{\Delta}{2\sqrt{\xi_k^2 + |\Delta|^2}} (f(E_\beta) - f(E_\alpha)) \quad (37)$$

Opomba, lahko definiramo tudi $n = n_a + n_b = \text{konst.} = 1$ in $n_a - n_b =: m$ od tod sledi

$$H^{\text{MF}} = \sum_k \left(\left[\epsilon_{a,k} + \frac{V}{2} n - \frac{V}{2} m \right] c_{a,k}^\dagger c_{a,k} + \left[\epsilon_{b,k} + \frac{V}{2} n - \frac{V}{2} m \right] c_{b,k}^\dagger c_{b,k} + \Delta^* c_{a,k}^\dagger c_{b,k} + \Delta c_{b,k}^\dagger c_{a,k} \right) \quad (38)$$

od koder je v 2D Hartree gap enak:

$$G^H = G - 4(t_a - t_b) - Vm. \quad (39)$$

Note: v praksi je $m < 0$ V dveh dimenzijah se Hartree pasova srečata na

$$\mu_B = \frac{V}{2} n - \frac{t_a + t_b}{t_a - t_b} \frac{G - Vm}{2}. \quad (40)$$

Zamik je enak $V/2$, ko sta osnovna pasova simetrična in je zasedenost polovična. V primeru simetričnih pasov sta disperziji kvazidelcev

$$E_{\alpha(\beta)}(k) - V/2 = \pm \sqrt{\left(\frac{\epsilon_{a,k} - \epsilon_{b,k} - Vm}{2}\right)^2 + |\Delta|^2}. \quad (41)$$

Velja še

$$n_a = \frac{1}{N} \sum_k (u_k^2 f(E_\alpha) + v_k^2 f(E_\beta)) \quad (42)$$

$$n_b = \frac{1}{N} \sum_k (v_k^2 f(E_\alpha) + u_k^2 f(E_\beta)) \quad (43)$$

$$u_k^2 = \frac{1}{2} \left(1 + \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right) \quad (44)$$

$$v_k^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right) \quad (45)$$

1.3 Numerika/psevdokoda

1.4 Odprti sistemi in modeliranje kopeli

Born-Markov master enačbe in Redfield: ločimo kopel in sistem,

$$\rho_S(t) = \text{Tr}_B(\rho(t)) \quad (46)$$

Hamiltonjan razcepimo na $H = H_0 + H_I$, kjer je $H_0 = H_S \otimes \mathbb{I}_B + \mathbb{I}_S \otimes H_B$ razdeljen na podсистema, za interakcijski del pa privzamemo obliko $H_I = \sum_\alpha A_\alpha \otimes B_\alpha$. V interakcijski sliki je operator za evolucijo

$$U_0(t, t_0) = e^{-iH_0(t-t_0)} \quad (47)$$

$$\rho_I(t) = U_0^\dagger(t, 0) \rho(t) U_0(t, 0) = e^{iH_0 t} \rho(t) e^{-iH_0 t} \quad (48)$$

$$H_I(t) = U_0^\dagger(t, 0) H_I U_0(t, 0) = e^{iH_0 t} H_I e^{-iH_0 t} \quad (49)$$

Von Neumannova enačba je

$$\frac{d}{dt} \rho_I(t) = -i[H_I(t), \rho_I(t)] \quad (50)$$

$$\rho_I(t) = \rho_I(0) - i \int_0^t dt' [H_I(t'), \rho_I(t')] \quad (51)$$

vstavimo nazaj v komutator in se znebimo prostosnih stopenj kopeli:

$$\frac{d}{dt} \rho_S(t) = - \int_0^t ds \text{Tr}_B ([H_I(t), [H_I(t-s), \rho_S(t) \otimes \rho_B]]) \quad (52)$$

V interakcijskem členu so operatorji A_α napisani kot lastni operatorji H_S z energijami prehodov $\omega = E_b - E_a$

$$A_\alpha(\omega) = \sum_{E_b - E_a = \omega} |a\rangle \langle a| A_\alpha |b\rangle \langle b| \quad (53)$$

To zadošča relacijam $[H_S, A_\alpha(\omega)] = -\omega A_\alpha(\omega)$, $[H_S, A_\alpha^\dagger(\omega)] = \omega A_\alpha^\dagger(\omega)$ in $e^{iH_S t} A_\alpha(\omega) e^{-iH_S t} = e^{-i\omega t} A_\alpha(\omega)$, $e^{iH_S t} A_\alpha^\dagger(\omega) e^{-iH_S t} = e^{i\omega t} A_\alpha^\dagger(\omega)$. V interakcijski sliki potem velja

$$H_I(t) = \sum_{\alpha, \omega} e^{-i\omega t} A_\alpha(\omega) \otimes B_\alpha(t) = e^{iH_B t} B_\alpha e^{-iH_B t} \quad (54)$$

evolucija sledi kot

$$\dot{\rho}_S = \sum_{\omega, \omega'} \sum_{\alpha, \beta} e^{i(\omega' - \omega)t} \Gamma_{\alpha\beta}(\omega) \left(A_\alpha(\omega) \rho_S(t) A_\beta^\dagger(\omega') - A_\beta^\dagger(\omega') A_\alpha(\omega) \rho_S(t) \right) + \text{h.c.} \quad (55)$$

kjer so prehodne matrike med stanji

$$\Gamma_{\alpha\beta}(\omega) = \int_0^\infty ds e^{i\omega s} \text{Tr}_B \left(B_\alpha^\dagger(t) B_\beta(t-s) \rho_B \right) = \frac{1}{2} \gamma_{\alpha\beta}(\omega) + i S_{\alpha\beta}(\omega) \quad (56)$$

Evolucija pa ima strukturo

$$\frac{d\rho_S}{dt} = -i[H_{LS}, \rho_S(t)] + \mathcal{D}(\rho_S(t)) \quad (57)$$

z disipatorjem

$$\mathcal{D}(\rho_S) = \sum_{\omega} \sum_{\alpha, \beta} \gamma_{\alpha\beta}(\omega) \left(A_\beta(\omega) \rho_S A_\alpha^\dagger(\omega) - \frac{1}{2} \{ A_\alpha^\dagger(\omega) A_\beta(\omega), \rho_S \} \right) \quad (58)$$

oz. v splošnem

$$\mathcal{D}(\rho_S) = \sum_k \gamma_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho_S \} \right) \quad (59)$$

V stacionarnem stanju velja

$$\mathcal{L}\rho_{\text{ness}} = 0. \quad (60)$$

Notes za vektorizacijo in superoperatorje:

$$\mathcal{L}u_\mu^R = \lambda_\mu u_\mu^R \quad (61)$$

$$\text{vec}(ABC) = (A \otimes C^T) \text{vec}(B) \quad (62)$$

$$\text{Tr}(A^\dagger B) = \text{vec}(A)^\dagger \text{vec}(B) \quad (63)$$

$$\frac{d}{dt} |\rho\rangle = \left(-iH \otimes \mathbb{I} + i\mathbb{I} \otimes H^T + \sum_k L_k \otimes L_k^* - \frac{1}{2} \sum_k (L_k^\dagger L_k \otimes \mathbb{I} + \mathbb{I} \otimes L_k^T L_k^*) \right) |\rho\rangle \quad (64)$$

$$\langle \hat{O} \rangle(t) = \langle \mathbb{I} | \hat{O} \otimes \mathbb{I} | \rho(t) \rangle \quad (65)$$

Šibke sklopitve in perturbacijski pristopi

$$\mathcal{L} = -i[H, \cdot] + \epsilon \mathcal{D}[\cdot], \quad \rho = \rho_0 + \epsilon \rho_1 + \epsilon^2 \rho_2 + \dots \quad (66)$$

napišemo gostotno matriko sistema v lastni bazi, siagonalno, s klasičnimi populacijami

$$\rho_0 = \sum_n p_n |n\rangle \langle n|, \quad H|n\rangle = E_n |n\rangle \quad (67)$$

kjer velja v ničtem redu (by construction)

$$\mathcal{L}\rho_0 = -i[H, \rho_0] = 0 \quad (68)$$

V prvem redu so enačbe za NESS

$$-i[H, \rho_1] + \mathcal{D}(\rho_0) = 0 \quad (69)$$

kjer je $\mathcal{D}(\rho_0) = \sum_{\mu} (L_{\mu}\rho_0 L_{\mu}^{\dagger} - \frac{1}{2}\{L_{\mu}^{\dagger}L_{\mu}, \rho_0\})$ in od tod enačba za prehode med populacijami

$$\langle n | \mathcal{D}(\rho_0) | n \rangle = \sum_{\mu} \sum_m p_m |\langle n | L_{\mu} | m \rangle|^2 - \sum_{\mu} \sum_m p_n |\langle m | L_{\mu} | n \rangle|^2 \quad (70)$$

kot klasična populacijska dinamika z $R_{nm} = \sum_{\mu} |\langle n | L_{\mu} | m \rangle|^2$

$$\dot{p}_n = \epsilon \sum_m (R_{nm} p_m - R_{mn} p_n) \quad (71)$$

Izven diagonalni elementi se razvijejo hitro na $\mathcal{O}(1)$, medtem ko se populacije razvijajo počasi na GGE manifoldu z $\mathcal{O}(\epsilon^{-1})$. Za ohranitev normalizacije mora veljati $\sum_n \dot{p}_n = 0$.

Integrabilni sistemi imajo makroskopsko število ohranjenih nabojev $[H_0, C_i] = 0$, pod unitarno časovno evolucijo se razvijejo v netermično stanje

$$\rho_{GGE} = \frac{1}{Z} e^{-\sum_i \lambda_i C_i} \quad (72)$$

$$\langle \psi_0 | C_i | \psi_0 \rangle = \text{Tr}(C_i \rho_{GGE}) \quad (73)$$

fiksira λ_i . Unitarna dinamika pripcira na GGE manifold, šibka sklopitev premika po GGE manifoldu:

$$\rho_0(t) = \rho_{\lambda(t)} = \frac{1}{Z(t)} \exp \left(- \sum_i \lambda_i(t) C_i \right) \quad (74)$$

EOM za ohranjene naboje je

$$\langle \dot{C}_i \rangle_{\lambda(\infty)} = \text{Tr}[C_i \mathcal{D}\rho] \simeq \epsilon \text{Tr}[C_i \mathcal{D}\rho_0] = 0 \quad \forall \lambda_i \quad (75)$$

Definiramo projektor na GGE manifold:

$$PX = - \sum_{ij} \frac{\partial \rho}{\partial \lambda_i} (\chi^{-1})_{ij} \text{Tr}[C_j X] \quad (76)$$

ki da dinamiko vzdolž manifolda:

$$\frac{\partial \rho_{\lambda}}{\partial t} = \sum_i \frac{\partial \rho_{\lambda}}{\partial \lambda_i} \dot{\lambda}_i = \epsilon P \mathcal{D}(\rho_{\lambda}) = -\epsilon \sum_{ij} \frac{\partial \rho_{\lambda}}{\partial \lambda_i} (\chi^{-1})_{ij} \text{Tr}[C_j \mathcal{D}(\rho_{\lambda})] \quad (77)$$

od koder sledi evolucija za koeficiente/potenciale λ_i :

$$\dot{\lambda}_i = -\epsilon \sum_j (\chi^{-1})_{ij} \text{Tr}[C_j \mathcal{D}\rho_{\lambda(t)}] \quad (78)$$

s susceptibilnostmi $\chi_{ij} = \langle C_i C_j \rangle - \langle C_i \rangle \langle C_j \rangle = -\text{Tr} \left[C_i \frac{\partial \rho}{\partial \lambda_j} \right]$.

2 Ekscitonski izolatorji in sklopitev s kopelmi

$$\text{Tr} \left[f \frac{1}{2} \{\rho, A\} \right] = \text{Tr} \left[\frac{1}{2} f \rho A + \frac{1}{2} f A \rho \right] = \frac{1}{2} \text{Tr} [A f \rho + f A \rho] \quad (79)$$

$$= \frac{1}{2} \text{Tr} [(A f + f A) \rho] = \frac{1}{2} \langle A f + f A \rangle \quad (80)$$

$$\dot{n}_k = \langle \dot{n}_k \rangle = \text{Tr}[n_k(t) \rho(t)] = -i \text{Tr}(n_k(t) [H(t), \rho(t)]) + \gamma_0 \text{Tr}[n_k(t) \mathcal{D}\rho] = 0 \quad (81)$$

$$2. \quad L_{qp}^{\lambda\lambda'} = \sqrt{\gamma_{qp}^{\lambda\lambda'}} d_{\lambda q}^\dagger d_{\lambda' p} \quad \text{for } (\lambda, \lambda') \in \{(\alpha\alpha), (\beta\beta), (\alpha\beta), (\beta\alpha)\} \quad (82)$$

$$\Rightarrow \quad \dot{n}_{\alpha k} = \text{Tr} [n_{\alpha k} \mathcal{D}(\rho)] = \sum_{\substack{q,p \\ \lambda, \lambda'}} \text{Tr} \left[n_{\alpha k} \left(L_{qp}^{\lambda\lambda'} \rho(t) L_{qp}^{\lambda\lambda'\dagger} - \frac{1}{2} \{ L_{qp}^{\lambda\lambda'\dagger} L_{qp}^{\lambda\lambda'}, \rho \} \right) \right] \quad (83)$$

$$= \sum_{\substack{q,p \\ \lambda, \lambda'}} \langle [L^\dagger, [n, L]] \rangle = ? \quad (84)$$

$$[n_{\alpha k}, d_{\lambda q}^\dagger d_{\lambda' p}] = d_{\lambda q}^\dagger [n_{\alpha k}, d_{\lambda' p}] + [n_{\alpha k}, d_{\lambda q}^\dagger] d_{\lambda' p} \quad (85)$$

$$\text{where } d_{\lambda q}^\dagger \{d_{\alpha k}^\dagger, d_{\lambda' p}\} - \{d_{\alpha k}^\dagger, d_{\lambda q}^\dagger\} d_{\lambda' p} = -\delta_{\alpha\lambda'} \delta_{kp} d_{\lambda' p} \quad (86)$$

$$= -d_{\lambda q}^\dagger d_{\lambda' p} \delta_{\alpha\lambda'} \delta_{kp} + d_{\lambda q}^\dagger d_{\lambda' p} \delta_{\alpha\lambda} \delta_{kq} \quad (87)$$

$$= \sum_{\substack{q,p \\ \lambda, \lambda' \\ T_1, T_2}} \left[\langle d_{\lambda' p}^\dagger d_{\lambda q} (-d_{\lambda q}^\dagger d_{\lambda' p}) \rangle \gamma_{qp}^{\lambda\lambda'} \delta_{\alpha\lambda'} \delta_{kp} + \langle d_{\lambda' p}^\dagger d_{\lambda q} d_{\lambda q}^\dagger d_{\lambda' p} \rangle \delta_{\alpha\lambda} \delta_{kq} \gamma_{qp}^{\lambda\lambda'} \right] \quad (88)$$

$$= \sum_{p, \lambda'} \Gamma_{kp}^{\alpha\lambda'} n_{\lambda' p} (1 - n_{\alpha k}) - \sum_{q, \lambda} \Gamma_{qk}^{\lambda\alpha} n_{\alpha k} (1 - n_{\lambda q}) \quad (89)$$

$$\dot{n}_{\alpha k} = \sum_{\substack{p, \lambda \\ \lambda \in \{\alpha, \beta\}}} \left[\Gamma_{kp}^{\alpha\lambda} n_{\lambda p} (1 - n_{\alpha k}) - \Gamma_{pk}^{\lambda\alpha} n_{\alpha k} (1 - n_{\lambda p}) \right] \quad (90)$$

2.1 Osnovni modeli kopeli

2.2 Limita neskončne temperature

2.3 Fononske sklopitve

$$\dot{n}_{\alpha k} = \sum_p \left(\Gamma_{\alpha p \rightarrow \alpha k} n_{\alpha p} (1 - n_{\alpha k}) - \Gamma_{\alpha k \rightarrow \alpha p} n_{\alpha k} (1 - n_{\alpha p}) + \right. \quad (91)$$

$$\left. + \Gamma_{\beta p \rightarrow \alpha k} n_{\beta p} (1 - n_{\alpha k}) - \Gamma_{\alpha k \rightarrow \beta p} n_{\alpha k} (1 - n_{\beta p}) \right) \quad (92)$$

$$\Gamma_{\mu p \rightarrow \nu k} = \frac{2\pi}{V} \sum_{\lambda} |G_{\mu\nu}^{\lambda}(k, q)|^2 \{ (1 + N_q) \delta(E_{\nu k} - E_{\mu p} + \omega_{q\lambda}) + N_{-q} \delta(E_{\nu k} - E_{\mu p} - \omega_{-q\lambda}) \} \quad (93)$$

$$\Gamma_{\nu k \rightarrow \mu p} = \frac{2\pi}{V} \sum_{\lambda} |G_{\mu\nu}^{\lambda}(k, q)|^2 \{ (1 + N_{-q}) \delta(E_{\mu p} - E_{\nu k} + \omega_{-q\lambda}) + N_q \delta(E_{\mu p} - E_{\nu k} - \omega_{q\lambda}) \} \quad (94)$$

$$G^{\lambda}(\mathbf{k}, \mathbf{q}) = \begin{pmatrix} u_{k+q} & -v_{k+q} \\ v_{k+q} & u_{k+q} \end{pmatrix} \begin{pmatrix} g_{a\lambda}(\mathbf{q}) & 0 \\ 0 & g_{b\lambda}(\mathbf{q}) \end{pmatrix} \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \quad (95)$$

$$g_{b\lambda}^a(\mathbf{q}) = -i \frac{4\pi e^2}{\sqrt{2M\Omega\omega_{q\lambda}}} \cdot \frac{1}{|\mathbf{q}|^2 + \Lambda^2} (\mathbf{q} \cdot \varepsilon_{\mathbf{q}\lambda}^*) \quad (96)$$

Psevdokoda:

2.4 Fotonske sklopitve v resonatorju

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Algorithm 1 pack: static bath data shared by all rate evaluations

Require: baths $\mathcal{B} = \{b\}$ with common $\{\mathbf{k}\}$, a_0 , n_{ang} , h (eh), E_{max} (em); $n_q = 512$

```
1:  $p \leftarrow 2\pi/a_0$  ▷ reciprocal period, per
2: check  $n_x n_y = N$ ,  $\Delta k_x = p/n_x$ ,  $\Delta k_y = p/n_y$  ▷ uniform Cartesian BZ grid
3: for  $b \in \mathcal{B}$  do
4:    $\text{kind}_b \leftarrow 0,1,2$  for constant, acoustic, gapped
5:    $\text{row}_b \leftarrow (T, c_s, k_{\text{TF}}, \eta, \text{amp}, c_{aa}, c_{ab}, c_{ba}, c_{bb}, q_D, \text{kind}, \omega_0, L, \epsilon_w)_b$ 
6:    $\text{tab}_b, \text{flat}_b \leftarrow$  angular kernel tables ▷ built, not used by the rates
7: end for
8:  $n_E \leftarrow \lceil E_{\text{max}}/h \rceil$ ,  $e_m \leftarrow m h$  for  $m = 0, \dots, n_E$ 
9:  $\tilde{E}_{\text{max}} \leftarrow e_{n_E} = n_E h$  ▷ stored as Pack.em
10: return  $P = (\{\mathbf{k}\}, \{\text{row}_b\}, p, \text{tab}, \text{flat}, \{e_m\}, h, \tilde{E}_{\text{max}})$ 
```

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Algorithm 2 Spectral and thermal factors (elementwise)

```

1: function SPEC $E(w, \omega, \epsilon)$  ▷  $\text{\_spec\_e}$ 
2:    $z_n \leftarrow \max(\frac{2}{\pi} \arctan(\omega/\epsilon), 10^{-30})$ 
3:   if  $\omega > 0$  then
4:     return  $\frac{4 \epsilon w \omega}{\pi [(w - \omega)^2 + \epsilon^2] [(w + \omega)^2 + \epsilon^2] z_n}$ 
5:   else
6:     return 0
7:   end if
8: end function
9: function SPEC $S(\omega, \epsilon)$  ▷  $\text{\_spec\_s}$ :  $\lim_{w \rightarrow 0} \text{SPEC}E/w$ 
10:   $z_n \leftarrow \max(\frac{2}{\pi} \arctan(\omega/\epsilon), 10^{-30})$ 
11:  if  $\omega > 0$  then
12:    return  $\frac{4 \epsilon \omega}{\pi (\omega^2 + \epsilon^2)^2 z_n}$ 
13:  else
14:    return 0
15:  end if
16: end function
17: function HAT $(x, h)$ 
18:   return  $\max(1 - |x|/h, 0)/h$ 
19: end function
20: function DISC $E(w, \omega, h)$  ▷  $\text{\_disc\_e}$ , used only if  $\epsilon_w = 0$ 
21:   $\rho \leftarrow \min(\omega/h, 1)$ ;  $z_n \leftarrow \rho(2 - \rho)$  if  $\omega < h$  else 1;  $z_n \leftarrow \max(z_n, 10^{-30})$ 
22:  if  $\omega > 0$  then
23:    return  $\max([\text{HAT}(w - \omega, h) - \text{HAT}(w + \omega, h)]/z_n, 0)$ 
24:  else
25:    return 0
26:  end if
27: end function
28: function DISC $S(\omega, h)$  ▷  $\text{\_disc\_s}$ 
29:   $z \leftarrow 10^{-6}h$ ;
30:  return DISC $E(z, \omega, h)/z$ 
31: end function
32: function LINE $(w, \omega, \epsilon_w, P)$  ▷  $\text{\_line}$ 
33:   $\epsilon_p \leftarrow \epsilon_w$  if  $\epsilon_w > 0$  else  $h$ 
34:  if  $\epsilon_w > 0$  then
35:     $A \leftarrow \text{SPEC}E(w, \omega, \epsilon_p)$ ;  $A_0 \leftarrow \text{SPEC}S(\omega, \epsilon_p)$ 
36:  else
37:     $A \leftarrow \text{DISC}E(w, \omega, h)$ ;  $A_0 \leftarrow \text{DISC}S(\omega, h)$ 
38:  end if
39:  if  $w > \tilde{E}_{\max}$  then  $A, A_0 \leftarrow \text{NaN}$ 
40:  end if
41:  return  $(A, A_0)$ 
42: end function
43: function THERM $(A, A_0, w, T)$  ▷  $\text{\_therm}$ 
44:  if  $T > 0$  then
45:     $x \leftarrow w/T$ ;  $d \leftarrow 1 - e^{-x}$  if  $x > 0$  else 1
46:     $F_{\text{abs}} \leftarrow A e^{-x}/d = A N(w)$  if  $w > 0$  else  $A_0 T$ 
47:  else
48:     $F_{\text{abs}} \leftarrow 0$ 
49:  end if
50:   $F_{\text{em}} \leftarrow F_{\text{abs}} + A = A [1 + N(w)]$ 
51:  return  $(F_{\text{abs}}, F_{\text{em}})$ 
52: end function

```

Algorithm 3 _data and _pair: pair rates between destinations i and sources j

```

1: function DATA(st,P) ▷ flattened states  $s = \lambda N + k, \lambda \in \{\alpha, \beta\}$ 
2:   for  $s = (\lambda, \mathbf{k})$  do
3:      $(a_s, b_s) \leftarrow (u_{\mathbf{k}}, -v_{\mathbf{k}})$  if  $\lambda = \alpha$ ;  $(v_{\mathbf{k}}, u_{\mathbf{k}})$  if  $\lambda = \beta$ 
4:      $x_s \leftarrow (E_{\lambda \mathbf{k}}, a_s, b_s, k_x, k_y)$ 
5:   end for
6:   return  $X = \{x_s\}$ 
7: end function
8: function PAIR( $X_i, X_j, P$ ) ▷ rows  $i$ : destinations, columns  $j$ : sources
9:    $\Delta E_{ij} \leftarrow E_i - E_j$ ;  $w_{ij} \leftarrow |\Delta E_{ij}|$ 
10:   $\mathbf{q}_{ij} \leftarrow \text{wrap}(\mathbf{k}_i - \mathbf{k}_j)$ ,  $\text{wrap}(x) = [(x + p/2) \bmod p] - p/2$  per component
11:   $q_{ij} \leftarrow |\mathbf{q}_{ij}|$ 
12:   $O_{ij}^{aa} \leftarrow a_i a_j$ ,  $O_{ij}^{ab} \leftarrow a_i b_j$ ,  $O_{ij}^{ba} \leftarrow b_i a_j$ ,  $O_{ij}^{bb} \leftarrow b_i b_j$ 
13:   $R^{(1)} \leftarrow 0$ ,  $R^{(2)} \leftarrow 0$ 
14:  for  $b \in \mathcal{B}$  do ▷ for i_loop over rows of  $P$ 
15:     $(T, c_s, k_{\text{TF}}, \eta, \text{amp}, c_{aa}, c_{ab}, c_{ba}, c_{bb}, q_D, \text{kind}, \omega_0, L, \epsilon_w) \leftarrow \text{row}_b$ 
16:     $\omega_g \leftarrow 2\pi/L$ 
17:     $\Omega_{ij} \leftarrow \begin{cases} \omega_0 & \text{kind} = 0 \\ c_s q_{ij} & \text{kind} = 1 \\ \sqrt{\omega_g^2 + c_s^2 q_{ij}^2} & \text{kind} = 2 \end{cases}$ 
18:     $\gamma_{ij} \leftarrow c_{aa} O_{ij}^{aa} + c_{ab} O_{ij}^{ab} + c_{ba} O_{ij}^{ba} + c_{bb} O_{ij}^{bb}$  ▷  $G_{\mu\nu}/g$ 
19:     $g_{ij}^2 \leftarrow \frac{\text{amp } \gamma_{ij}^2 q_{ij}^2}{\max(\Omega_{ij}, 10^{-30}) (q_{ij}^2 + k_{\text{TF}}^2)^2}$ 
20:     $M_{ij} \leftarrow 2\pi g_{ij}^2$  if  $0 < q_{ij} \leq q_D$  else 0
21:     $(A, A_0) \leftarrow \text{LINE}(w_{ij}, \Omega_{ij}, \epsilon_w, P)$ 
22:     $(F_{\text{abs}}, F_{\text{em}}) \leftarrow \text{THERM}(A, A_0, w_{ij}, T)$ 
23:     $R_{ij}^{(1)} \leftarrow R_{ij}^{(1)} + M_{ij} \cdot (F_{\text{em}} \text{ if } \Delta E_{ij} < 0 \text{ else } F_{\text{abs}})$  ▷ rate  $j \rightarrow i$ 
24:     $R_{ij}^{(2)} \leftarrow R_{ij}^{(2)} + M_{ij} \cdot (F_{\text{em}} \text{ if } \Delta E_{ij} > 0 \text{ else } F_{\text{abs}})$  ▷ rate  $i \rightarrow j$ 
25:  end for
26:  return  $(R^{(1)}, R^{(2)})$ 
27: end function

```

Algorithm 4 rates, source_rates and cur_block

```
1: function RATES(st, P, J) ▷ J: source columns, default all
2:   X ← DATA(st, P)
3:   (R(1), ·) ← PAIR(X, XJ, P)
4:   Rij ← Rij(1) if i ≠ j else 0 ▷ R[dest,src] = Γj→i
5:   return R
6: end function
7: function SOURCERATES(st, B, μ, p0)
8:   P ← PACK(B)
9:   return RATES(st, P, {μN + p0}) reshaped to (2, N) ▷ Γμp0→νk
10: end function
11: function CURBLOCK(st, n, w, P, B) ▷ wk: k-point weights, B: block size
12:   X ← DATA(st, P);   ys ← ns;   w̄s ← wk(s) ▷ weights tiled over both bands
13:   gain ← 0,   loss ← 0
14:   for each block J of B source indices (zero-padded) do
15:     (R(1), R(2)) ← PAIR(X, XJ, P)
16:     set Rij(1) = Rij(2) = 0 for i = j and for padded j
17:     gaini += ∑j∈J Rij(1) w̄j yj
18:     lossi += ∑j∈J Rij(2) w̄j (1 - yj)
19:   end for
20:   ṅi ← (1 - yi) gaini - yi lossi
21:   return ṅ reshaped to (2, N)
22: end function
```

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